Z. 2026 **Concurrency Theory** P. Hofman

Tutorial 2, Friday, 09/10/2026, by VRM

Today: *Büchi Automata & ω-regular languages*

A. Automata and Expressions for Infinite Words

Automata on ω-words. A *nondeterministic automaton* is a tuple

$$\mathcal{A} = (Q, \Sigma, \delta, q_0, \mathcal{F}),$$

where Q is a finite set of states, Σ is a finite alphabet, $\delta \subseteq Q \times \Sigma \times Q$ is the transition relation, $q_0 \in Q$ is the initial state, and $\mathcal{F}$ is some sort of *acceptance condition* on words of Σ^ω; in *deterministic* automata, $\delta : Q \times \Sigma \to Q$ is a (complete/total) function. For a run

$$\rho : q_0 \xrightarrow{a_0} q_1 \xrightarrow{a_1} q_2 \xrightarrow{a_2} \cdots$$

we shall write

$$\mathrm{Inf}(\rho) = \{q \in Q : q = q_i \text{ for infinitely many } i\}$$

for the set of states occurring infinitely often in ρ.

Büchi automata. In a *Büchi automaton* (NBA), the acceptance condition is generated by a set $F \subseteq Q$ of states, and a run ρ is *accepting* if it visits F infinitely often: $\mathrm{Inf}(\rho) \cap F \neq \emptyset$.

Other automata. There are other automata that are similar to Büchi, with different acceptance conditions.

- A *Muller automaton* is equipped with an acceptance family $\mathcal{F} \subseteq 2^Q$. A run ρ is accepting if the set of infinitely occurring states is an element of the family $\mathcal{F}$: in symbols, $\mathrm{Inf}(\rho) \in \mathcal{F}$.
- A *Rabin automaton* is an automaton equipped with an acceptance condition

  $$\mathcal{R} = \{(E_1, F_1), \ldots, (E_k, F_k)\}, \qquad E_i, F_i \subseteq Q.$$

  A run ρ is accepting if there exists $i \in \{1, \ldots, k\}$ such that

  $$\mathrm{Inf}(\rho) \cap E_i = \emptyset \quad \text{and} \quad \mathrm{Inf}(\rho) \cap F_i \neq \emptyset,$$

  i.e. if it visits an "allowed" set E_i infinitely often, and avoids the corresponding "forbidded" set co-finitely often (co-finitely: always, except a finite number of times).
- A *parity automaton* is an automaton equipped with a priority function $\Omega : Q \longrightarrow \{0, \ldots, d\}$, and a run ρ is accepting if

  $$\min\{\Omega(q) : q \in \mathrm{Inf}(\rho)\}$$

  is even.

ω-regular Languages. An *ω-regular expression* over an alphabet Σ is defined inductively by the grammar

$$s ::= r^\omega \mid rs \mid s + s,$$

where r ranges over *regular expressions* over Σ. Its language $\mathcal{L}(s) \subseteq \Sigma^\omega$ is defined inductively by

$$\begin{aligned} \mathcal{L}(r^\omega) &= \mathcal{L}(r)^\omega, \\ \mathcal{L}(rs) &= \mathcal{L}(r)\mathcal{L}(s), \\ \mathcal{L}(s_1 + s_2) &= \mathcal{L}(s_1) \cup \mathcal{L}(s_2). \end{aligned}$$

A language $L \subseteq \Sigma^\omega$ is called *ω-regular* if $L = \mathcal{L}(s)$ for some ω-regular expression s.

McNaughton's determinization theorem. It is a complicated theorem[1] that every nondeterministic Büchi automaton is *equivalent to* (i.e., it recognizes the same language as) a *deterministic* Muller automaton.

Among other results, this allows us to prove the closure of ω-regular languages under complementation: if $L \subseteq \Sigma^\omega$ is ω-regular, then $\Sigma^\omega - L$ is also ω-regular.

Proof. By Büchi's characterization, there exists a nondeterministic Büchi automaton $\mathcal{A}$ recognizing L. By McNaughton's theorem, there exists an equivalent deterministic Muller automaton $\mathcal{B}$. Complementing the Muller acceptance family of $\mathcal{B}$ produces a deterministic Muller automaton recognizing $\Sigma^\omega - L$. Finally, every language recognized by a Muller automaton is recognized by a nondeterministic Büchi automaton. Hence $\Sigma^\omega - L$ is ω-regular. □

Usual Automata Problems. The fundamental decision problems for Büchi automata[2] are:

1. *Emptiness:* given $\mathcal{A}$, decide whether $\mathcal{L}_\omega(\mathcal{A}) = \varnothing$.
2. *Disjointness:* given $\mathcal{A}, \mathcal{B}$, decide whether $\mathcal{L}_\omega(\mathcal{A}) \cap \mathcal{L}_\omega(\mathcal{B}) = \varnothing$.
3. *Universality:* given $\mathcal{A}$, decide whether $\mathcal{L}_\omega(\mathcal{A}) = \Sigma^\omega$.

B. Exercises

Exercise 1. [*Constructing Büchi automata*] Let $\Sigma = \{a, b\}$. Construct Büchi automata recognizing the following languages.

1. Words containing infinitely many occurrences of a.
2. Words containing only finitely many occurrences of a.
3. Words containing infinitely many occurrences of both a and b.
4. Words in which every occurrence of a is eventually followed by an occurrence of b.
5. Words containing infinitely many a's, but no two consecutive occurrences of a.
6. Words containing infinitely many b such that, between every two *consecutive* b's, the number of a's is divisible by 3.
7. Words such that, between every two consecutive b's, the number of a's is divisible by 3. There is no requirement that b occurs infinitely often.

[1] Proven in JAiO II.
[2] ...and all kinds of automata, in general!

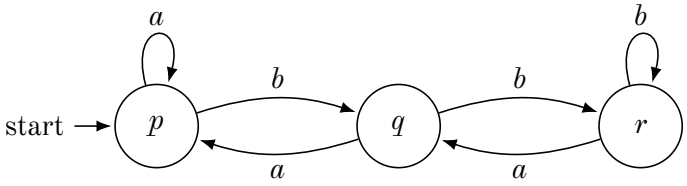
8. The number of b's is infinite, and only finitely many gaps between consecutive b's contain a number of a's divisible by 3. [*Hint*: equivalently, from some point on, every gap between consecutive b's has length non-zero modulo 3.]

9. The number of b's is finite, and every gap between consecutive b's contains a number of a's divisible by 3.

10. The number of b's is finite, and for every two *distinct* occurrences of b, the number of a's between them is not divisible by 3. [*Hint*: how many b's can there be?]

Which of these languages can be recognized by deterministic Büchi automata?

Exercise 2. [*Acceptance conditions*] Let $Q = \{p, q, r\}$. Describe which runs are accepted by the depicted automaton with each of the following conditions.



(a) The Muller condition $\mathcal{F} = \{\{p\}, \{p, q\}, \{q, r\}\}$.

(b) The Rabin condition $\mathcal{R} = \{(\{p\}, \{q\}), (\{q\}, \{r\})\}$.

(c) The parity condition $\Omega(p) = 0, \qquad \Omega(q) = 1, \qquad \Omega(r) = 2.$

Give a Muller condition equivalent to the Rabin condition in (b). Give a Rabin condition equivalent to the parity condition in (c).

Exercise 3. [*Pumping lemma for Büchi automata*] State and prove a pumping lemma for Büchi automata. Use it to prove that the following languages are not ω-regular.

$$L_1 = \{a^n b^n c^\omega : n \geq 1\}.$$
$$L_2 = \{a^n b^{2n} (ab)^\omega : n \geq 1\}.$$
$$L_3 = \{a^n b^m c^\omega : n, m \geq 1,\ n < m < 2n\}.$$
$$L_4 = \{a^{n_1} b a^{n_2} b a^{n_3} b \cdots : n_i = n_1 \text{ for every } i \geq 1\}.$$

Exercise 4. Are nondeterministic Büchi automata equivalent to *deterministic* ones? Find a proof of determinization, or a counterexample (with proof of impossibility).

Exercise 5. [*Complementing deterministic Muller automata*] Let $\mathcal{A} = (Q, \Sigma, \delta, q_0, \mathcal{F})$ be a deterministic Muller automaton, where $\delta : Q \times \Sigma \to Q$ is a (complete) function. Define $\overline{\mathcal{F}} = 2^Q - \mathcal{F}$. Prove that the Muller automaton $\overline{\mathcal{A}} = (Q, \Sigma, \delta, q_0, \overline{\mathcal{F}})$ recognizes the complement of $\mathcal{L}_\omega(\mathcal{A})$.

Do a similar construction for Rabin and parity automata.

Exercise 6. [*Büchi acceptance as other acceptance conditions*] Show that any Büchi automata can be simulated by Muller, Rabin, and parity automata.

Exercise 7. [*Parity to Rabin*] Let $\mathcal{A}$ be a parity automaton with priority function $\Omega : Q \to \{0, \ldots, d\}$. For every even priority $2i \leq d$, define

$$E_i = \{q : \Omega(q) < 2i\}, \qquad F_i = \{q : \Omega(q) = 2i\}.$$

Prove that the Rabin condition consisting of the pairs (E_i, F_i) is equivalent to the original parity condition of $\mathcal{A}$.

Exercise 8. [*Rabin to Muller*] Let $\mathcal{R} = \{(E_i, F_i) : 1 \leq i \leq k\}$ be a Rabin acceptance condition. Define

$$\mathcal{F} = \{S \subseteq Q \ : \ \exists i \ (S \cap E_i = \varnothing \ \wedge \ S \cap F_i \neq \varnothing)\}.$$

Prove that the Muller condition $\mathcal{F}$ is equivalent to $\mathcal{R}$.

Exercise 9. [*Rabin to Büchi*] Show that every language recognized by a nondeterministic Rabin automaton is recognized by a nondeterministic Büchi automaton. [*Hint*: Nondeterministically choose a Rabin pair (E_i, F_i) and guess a position after which no state from E_i is visited. From that position onward, require infinitely many visits to F_i.]

Exercise 10. [*Muller to Büchi*] Show that every language recognized by a nondeterministic Muller automaton is recognized by a nondeterministic Büchi automaton. [*Hint*: Nondeterministically select $S \in \mathcal{F}$, guess a position after which the run visits only states of S, and ensure that every state of S occurs infinitely often.]

Exercise 11. [*Büchi's characterization*] Prove that for every $L \subseteq \Sigma^\omega$, the following are equivalent:

1. L is ω-regular.
2. L is recognized by a nondeterministic Büchi automaton.

[(2)⇒(1)]: Let $U, V \subseteq \Sigma^*$ be regular languages, with $\varepsilon \notin V$. Construct a Büchi automaton recognizing UV^ω by combining the finite automata for U and V. The Büchi automaton should read a prefix in U, then repeatedly read nonempty words in V, visiting an accepting state after each repetition. Explain how to extend the construction to finite unions, and why this completes the proof of this direction.

[(1)⇒(2)]: Let $\mathcal{A} = (Q, \Sigma, \delta, q_0, F)$ be a Büchi automaton. For $p, q \in Q$, define

$$L_{p,q} = \{u \in \Sigma^* : \text{there exists a path from } p \text{ to } q \text{ labeled } u\}.$$

Observe that each $L_{p,q}$ is regular. Prove that

$$\mathcal{L}_\omega(\mathcal{A}) = \bigcup_{q \in F} L_{q_0,q} \left(L_{q,q} - \{\varepsilon\}\right)^\omega.$$

Exercise 12. [*Emptiness*] Show that the Emptiness problem for Büchi automata is decidable. What is the complexity of that problem?

Exercise 13. [*Intersection*] Let $\mathcal{A}_1, \mathcal{A}_2$ be Büchi automata. Prove that there exists a Büchi automaton $\mathcal{B}$ such that

$$\mathcal{L}_\omega(\mathcal{B}) = \mathcal{L}_\omega(\mathcal{A}_1) \cap \mathcal{L}_\omega(\mathcal{A}_2).$$

[*Hint*: The naive product with accepting set $F_1 \times F_2$ does not work for regular Büchi automata.] Deduce that the Disjointness problem is decidable. (*Optional*: What is its complexity?)

Exercise 14. Prove that the class of ω-regular languages is closed under union, intersection, and complement. Deduce that Universality is decidable.

Exercise 15. [*Finite prefixes*] Let $L \subseteq \Sigma^\omega$ be ω-regular and $K \subseteq \Sigma^*$ be regular. Prove that

$$KL = \{uv \mid u \in K, \ v \in L\}$$

is also ω-regular.

Exercise 16. [*Prefix independence*] Call an ω-language L *prefix-independent* if

$$uw \in L \iff w \in L$$

for every $u \in \Sigma^*$ and $w \in \Sigma^\omega$.

1. Prove that the following languages are prefix-independent:
   (a) words containing infinitely many a's;
   (b) words containing finitely many a's;
   (c) words containing infinitely many occurrences of ab.
2. Give an example of an ω-regular language that is *not* prefix independent.

Exercise 17. [*Ultimately periodic words*] Let

$$\mathrm{UP} = \{uv^\omega : u, v \in \Sigma^*,\ v \neq \varepsilon\}$$

be the set of *ultimately periodic words*.

Show that every nonempty language recognized by a Büchi automaton contains an ultimately periodic word. Use this fact to prove that the following languages are not ω-regular.

1. The language of infinite words over $\{a, b\}$ containing infinitely many b's and arbitrarily long consecutive blocks of a's.
2. $$L = \{a^{n_1} b a^{n_2} b a^{n_3} b \cdots : n_1 < n_2 < n_3 < \cdots\}.$$

Prove that two ω-regular languages $L_1, L_2 \subseteq \Sigma^\omega$ are equal if and only if

$$L_1 \cap \mathrm{UP} = L_2 \cap \mathrm{UP}.$$

[*Hint*: Look at the symmetric difference.]

Exercise 18. [*Myhill–Nerode for ω-languages*] By examining the behavior of the language

$$L = \left\{ \alpha \in \{a, b\}^* \;\middle|\; \begin{array}{l} \alpha \text{ contains infinitely many } b\text{'s, and} \\ \text{arbitrarily long blocks of } a\text{'s} \end{array} \right\}$$

under all three relations below, or otherwise, determine which of the following equivalence relations $\sim_L$ are candidates for a Myhill-Nerode theorem, like this: for $L \subseteq \Sigma^\omega$,

$$L \text{ is } \omega\text{-regular} \quad \iff \quad \sim_L \text{ has finite index.}$$

1. The equivalence relation on Σ^* defined by
   $$u \sim_L v \iff \forall w \in \Sigma^\omega : (uw \in L \iff vw \in L).$$
2. The equivalence relation on Σ^ω defined by
   $$u \sim_L v \iff \forall w \in \Sigma^* : (wu \in L \iff wv \in L).$$
3. The equivalence relation on Σ^* defined by
   $$u \sim_L v \iff \begin{cases} uw \in L \iff vw \in L & \text{for all } w \in \Sigma^\omega, \\ s(ut)^\omega \in L \iff s(vt)^\omega \in L & \text{for all } s, t \in \Sigma^* \end{cases}$$
   where the second condition is understood only when $ut, vt \neq \varepsilon$.

Which implications still work, and which fail?

Exercise 19. [*co-Büchi automata*] A *co-Büchi automaton* is an automaton $\mathcal{A} = (Q, \Sigma, \delta, q_0, F)$ in which a run ρ is accepting if

$$\mathrm{Inf}(\rho) \subseteq F,$$

i.e., if it visits states outside F only finitely often. The aim of this exercise is to prove that every nondeterministic co-Büchi automaton is equivalent to a deterministic co-Büchi automaton.

Adapt the subset construction, using as states pairs (S, R) with $R \subseteq S \subseteq Q$. The set S tracks all states reachable after reading the current prefix, whereas R tracks those reachable by runs that have remained inside F since the last time R was empty.

Initially, $(S, R) = (\{q_0\}, \varnothing)$. On reading a letter a, update

$$S' = \delta(S, a), \qquad R' = \begin{cases} \delta(S, a) \cap F, & \text{if } R = \varnothing, \\ \delta(R, a) \cap F, & \text{if } R \neq \varnothing. \end{cases}$$

Equip the resulting deterministic automaton with the co-Büchi accepting set

$$\{(S, R) : R \neq \varnothing\}.$$

Prove that it recognizes the same language as $\mathcal{A}$.

C. Summary

Here is a summary of properties of finite-word and ω-automata. An underlined Y indicates that the corresponding construction or algorithm requires only polynomial time. "RegEx equiv." means that the model is equivalent to (ω-)regular expressions, "Det." indicates whether for every nondeterministc automaton there is an equivalent deterministic one with the same acceptance condition, and the other three columns indicate whether the corresponding operation can be performed **without extensive modification** of the respective transition graphs.

Automaton type		RegEx equiv.	Det.	Union	Inters.	Comp.
NFA/DFA		Y	Y	Y	Y	Y
NBA/DBA	(Büchi)	Y	N	Y	N	N
NCA/DCA	(Co-Büchi)	N	Y	N	Y	N
NRA/DRA	(Rabin)	Y	Y	Y	N	N
NSA/DSA	(Streett)	Y	Y	N	Y	N
NPA/DPA	(Parity)	Y	Y	N	N	Y
NMA/DMA	(Muller)	Y	Y	Y	Y	Y